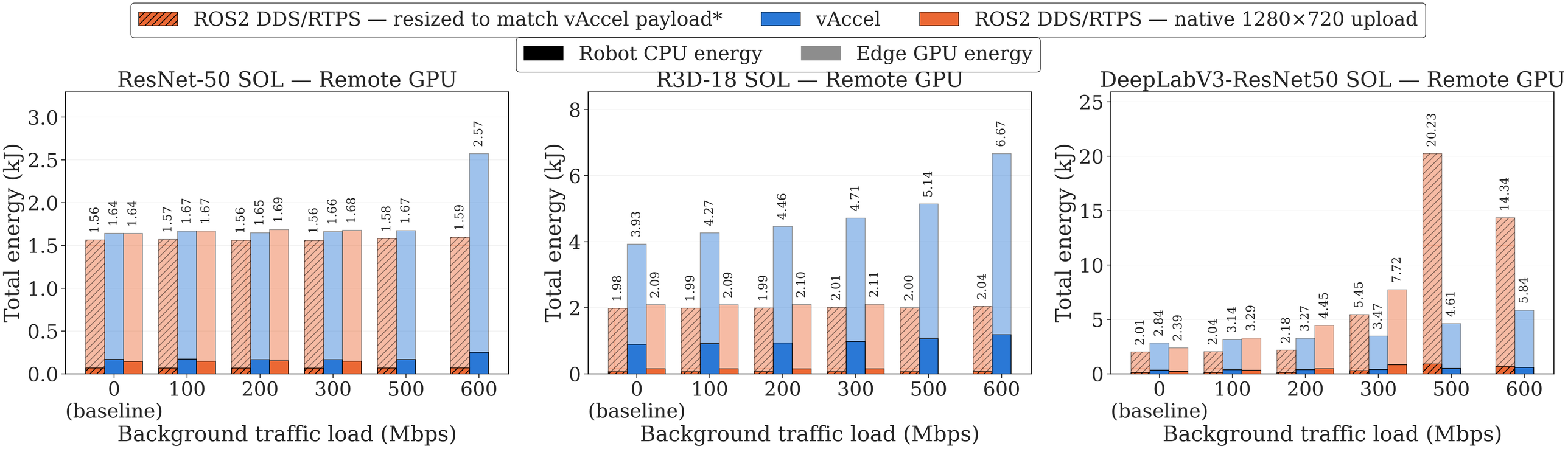
ROS2 DDS/RTPS — resized to match vAccel payload*
vAccel
ROS2 DDS/RTPS — native 1280×720 upload
Robot CPU energy
Edge GPU energy
ResNet-50 SOL — Remote GPU
R3D-18 SOL — Remote GPU
DeepLabV3-ResNet50 SOL — Remote GPU
Total energy (kJ)
Background traffic load (Mbps)
0 (baseline)
100
200
300
500
600
1.56 1.64 1.64
1.57 1.67 1.67
1.56 1.65 1.69
1.56 1.66 1.68
1.58 1.67
1.59 2.57
1.98 3.93 2.09
1.99 4.27 2.09
1.99 4.46 2.10
2.01 4.71 2.11
2.00 5.14
2.04 6.67
2.01 2.84 2.39
2.04 3.14 3.29
2.18 3.27 4.45
5.45 3.47 7.72
20.23 4.61
14.34 5.84